

Validation of a Pharmacist Monitoring Tool for Patients with Inflammatory Bowel Disease

Connie Vo, PharmD, CSP¹; Kristina M. Heimerl, PharmD, BCACP¹
Jaynika B. Patel, PharmD¹ James Langley, PharmD, MS¹

BACKGROUND

- Monitoring specialty medications utilized for treating inflammatory bowel diseases (IBD) is important due to their potential to cause adverse effects and need for dose adjustments to ensure optimal response
- Validated IBD physician assessment tools standardize assessment of disease state activity, but do not address medication-related concerns which lead to lower applicability of the tools for pharmacists
- In 2019 at University of Wisconsin (UW) Health, a pharmacist IBD patient monitoring tool, the gastro-intestinal metric (GIM) Score, was developed in collaboration with payors, patients, physicians, and pharmacists to standardize assessments and identify suboptimal medication efficacy

Gastro-Intestinal Metric (GIM) Score

Assessment Question Category	
General well-being	Disease state progression
Abdominal pain	
Bowel frequency	
Blood in stool	
Resolution of symptom attempts	Medication efficacy
Work/school productivity	
Next dose awareness	
Missed doses	
Medication adverse effects	

- The GIM Score is currently being utilized by the specialty pharmacy team during routine telephone follow-ups with patients taking IBD specialty medications

OBJECTIVES

Validate the GIM Score through comparison of pharmacist and physician assessments of patients' IBD control at the time of office visit (baseline)

Measure the accuracy of the GIM Score to develop a cutoff score for IBD stability and responsiveness of the GIM Score for patients with changing disease activity

A validated pharmacist-specific IBD patient monitoring tool can **standardize pharmacist assessments**, allowing for a **consistent approach** to recommending medication optimization and disease state follow-up

METHODS

Type: Prospective cohort study

Time frame: August 1, 2024 to July 31, 2025

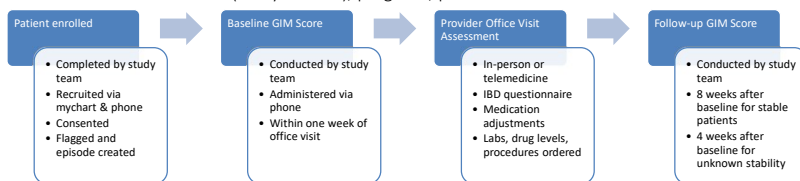
IRB: Minimal risk research approval

Study team: specialty pharmacists, student researchers, biostatistician

Inclusion criteria: Established with a UW Health gastroenterologist (GI) and has an office visit scheduled during the study period; gastroenterologist is the prescriber of the patient's adalimumab (or adalimumab biosimilar), certolizumab, ustekinumab, risankizumab, tofacitinib, upadacitinib or ozanimod; adherent to IBD medication (PDC>0.7); participant in UW Health Specialty Pharmacy Medication Management Program

Exclusion criteria: Pediatrics (<18 years old), pregnant, prisoners

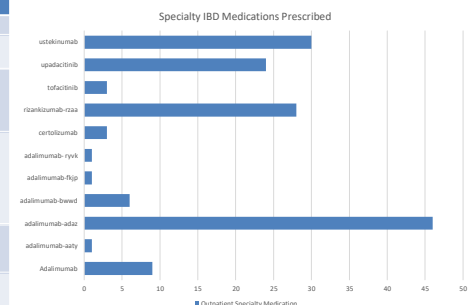
Study Design & Workflow



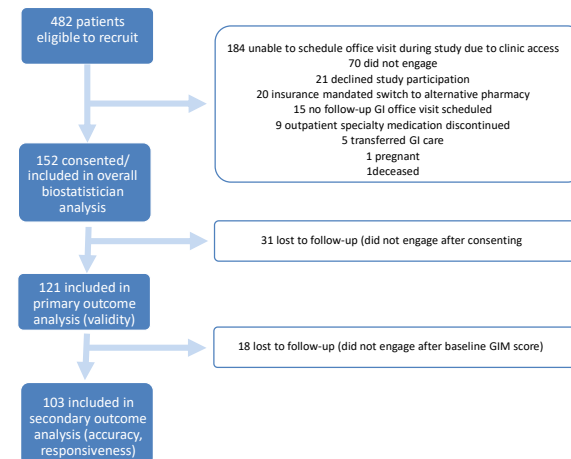
RESULTS

Table 1. Demographic Information (n=152)

Variables	Value (%)
Age	
median (range) years	48 (19-82)
Sex	
Female	80 (52.6)
Male	71 (46.7)
Non-binary	1 (0.7)
Race	
White	140 (92.1)
African American	5 (3.3)
Asian	4 (2.6)
Unknown	2 (1.3)
Indian or Alaska Native	1 (0.7)
IBD diagnosis	
Crohn's disease	112 (73.7)
Ulcerative colitis	40 (26.3)
Severe disease	
J-pouch or ostomy	22 (14.5)



RESULTS IN PROGRESS



DISCUSSION

- Integration of recruitment processes into specialty pharmacy workflows supports patient enrollment
- Collaborating with gastroenterology clinic providers and staff can increase visibility of specialty pharmacy research
- Limited gastroenterology clinic access can be a barrier to gathering clinic visit data
- Patient contact methods in addition to telephone, electronic health record messaging and letters are needed to support engagement

REFERENCES

- Cai Z, Wang S, Li J. Treatment of Inflammatory Bowel Disease: A Comprehensive Review. *Front Med (Lausanne)*. 2021;8:765474.
- Harvey RF, Bradshaw JM. A simple index of Crohn's-disease activity. *Lancet*. 1980;1(8167):514.
- Best WR, Becktel JM, Singleton JW. Rederived values of the eight coefficients of the Crohn's Disease Activity Index (CDAI). *Gastroenterology*. 1979;77(4 Pt 2):843-6.
- Lewis JD, Chuai S, Nessel L, Lichtenstein GR, Aberra FN, Ellenberg JH. Use of the noninvasive components of the Mayo score to assess clinical response in ulcerative colitis. *Inflamm Bowel Dis*. 2008;14(12):1660-6.
- Walsley RS, Ayres RC, Pounder RE, Allan RN. A simple clinical colitis activity index. *Gut*. 1998;43(1):29-32.

Affiliations

- Department of Pharmacy, UW Hospitals and Clinics; Madison, Wisconsin